

Karnataka Gravity Depth Diagnosis

Analysis of the sparse inversion run on AWS EC2: data fit, the recovered density model, why the model extends to the bottom of the mesh, and a comparison of α_s , norms, depth weighting and L1–L2.

AREA

70 × 70 km, easting 641–711 km, northing 1634–1704 km (UTM 43N, EPSG:32643)

MESH

1 km × 1 km × 500 m core, 10 km deep, 20 km padding (factor 1.3): 84 × 84 × 27 = 190,512 cells

DATA

NGPM Bouguer anomaly, 500 m grid decimated to 1 km: 5040 points; second-order trend surface removed

RUNS

AWS EC2 ap-south-1: sparse on c5.2xlarge (4.6 min), L1–L2 on c5.4xlarge (3.5 min); 15-run parameter study run locally; 4 checks on the 1 km mesh on 28 September (2 submitted from the upload page)

Summary

- **The data fit is fine.** $\chi^2/N = 0.97$, residual RMS 0.49 mGal, correlation between observed and predicted data 0.9986.
- **The model has no vertical resolution.** Below the core area the share of |mass| per 500 m layer barely changes from the surface to 10 km (2.9% $\rightarrow$ 2.3%); 50% of the anomalous mass lies in the padding below 10 km (D50 = 9.1 km). The power spectrum of the residual anomaly puts the sources at 1.1, 2.2 and 6.0 km.
- **The main cause is $\alpha_s = 10^{-4}$.** The pipeline passes $\alpha_{x,y,z}$ as length scales, so SimPEG weights the gradient terms by $(1 \times 500 \text{ m})^2$; per cell the smallness term is about 2500 times weaker, and its compactness ($p_s = 0$) has almost no effect. With an L1 vertical gradient ($p_z = 1$), a column that is constant with depth costs almost nothing. Changing only the depth weighting, with α_s unchanged, leaves the result as it was.
- **$\alpha_s = 1$ with Li & Oldenburg depth weighting removes the columns.** On the 2 km mesh ($\beta = 1$), the share of mass inside the core rises from 25% to 80%, the lateral padding share falls from 26% to 5%, D50 = 4.6 km, and the fit is unchanged. A smaller β gives a shallower model; the classic $\beta = 2$ is deeper here.
- **L1–L2 is equally compact.** At full resolution on EC2: 80% of the mass in the core, D50 = 3.0 km, strongest positive body (north-west) at 0.75–2.25 km.
- **The full-resolution check confirms this.** On the 1 km mesh with $\alpha_s = 1$, the dense body under the main Bouguer high lies at 2.3–4.8 / 3.3–7.8 / 4.3–11.1 km for $\beta = 0.5 / 1 / 1.5$, with nearly the same fit (χ^2/N 0.86–0.96). $\beta = 0.5$ and L1–L2 give the same depth. At the same place, the 2 km study predicted the base of the body to within 0.5 km (Section 7).
- **Caveat.** Gravity data alone do not determine depth; the regularization does. "Shallower" here means the model is no longer pushed to the bottom of the mesh. The actual depths need constraints from rock densities, geological sections or seismic data.

1 Data and preprocessing

Within the area the Bouguer anomaly ranges from -94.9 to -44.3 mGal. Its main feature is a NW–SE trending high from (650, 1685) to (675, 1660) km. The second-order trend surface (regional field) ranges from -95.3 to -75.9 mGal; removing it lowers the standard deviation of the data from 9.99 to 9.35 mGal and leaves a residual of -16.4 to +31.7 mGal. Every point was given an error of 0.5 mGal (noise floor).

Sign convention. Bouguer data are positive downward (positive over dense rock), while SimPEG's gz is positive upward. Before these runs the pipeline was fixed accordingly: the data are multiplied by -1 before SimPEG, and the observed and predicted data in the result are returned in the original convention. Without the fix, field data invert to a density model of the wrong sign.

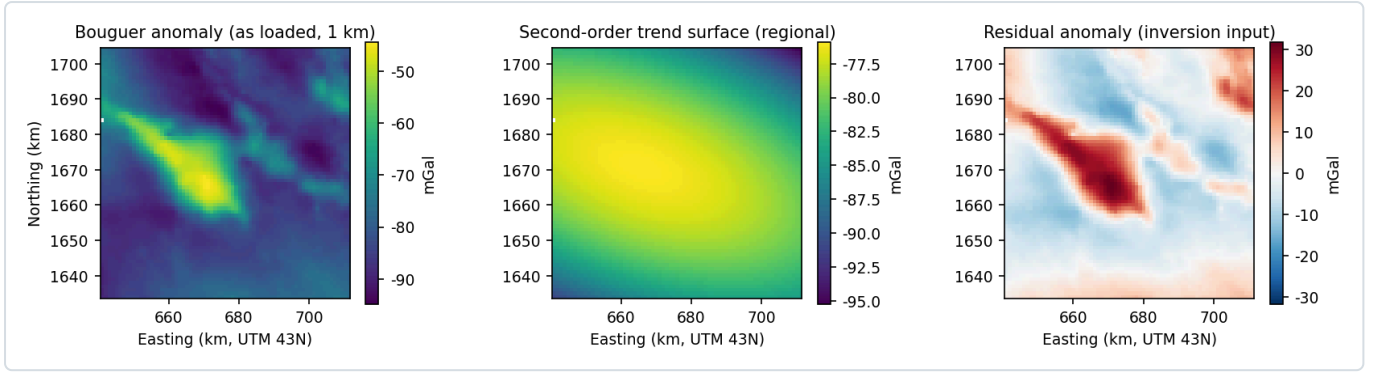


Figure 1 Left: Bouguer anomaly in the area (decimated to 1 km). Centre: least-squares second-order trend surface, removed as the regional field. Right: the residual anomaly, i.e. the data inverted (red is positive, dense).

2 Inversion settings

Regularization	sparse (IRLS), norms $p = [p_s, p_x, p_y, p_z] = [0, 2, 2, 1]$
Weights	$\alpha_s = 10^{-4}$, $\alpha_x = \alpha_y = \alpha_z = 1$ (length scales)
Depth weighting	SimPEG sensitivity weighting (UpdateSensitivityWeights)
Bounds	-0.2 to +0.5 g/cc density contrast (rock samples 2.52–3.40 g/cc, median 2.66)
Choice of β	$\chi^2 = N$ (DampedUpdateIRLS); at most 30 iterations and 30 IRLS iterations
Mesh	$84 \times 84 \times 27$ cells with a $70 \times 70 \times 20$ core; the 7 bottom padding layers are 650 m to 3.1 km thick, and the mesh ends at about 22 km depth

3 Convergence and data fit

The target $\varphi_d = N = 5040$ was reached at iteration 16; the following IRLS iterations kept φ_d between 4770 and 4895, ending at $\varphi_d = 4895$ ($\chi^2/N = 0.97$). During iterations 5–15 φ_d stayed near 5500 with β unchanged while IRLS adjusted the sparsity weights.

Residual RMS is 0.49 mGal, the largest residual 7.7 mGal, and 5.8% of the points have $|\text{residual} / \sigma| > 2$. The normalized residuals are **not Gaussian**: a sharp central peak with long tails. Most smooth areas are fitted better than the error level (over-fitting), while isolated spikes remain (single points in the residual map, Figure 3), most likely outliers from the original gridding. We suggest removing them, or using a robust (L1) misfit, before the next run; a 0.5 mGal noise floor may also be too small for gridded data.

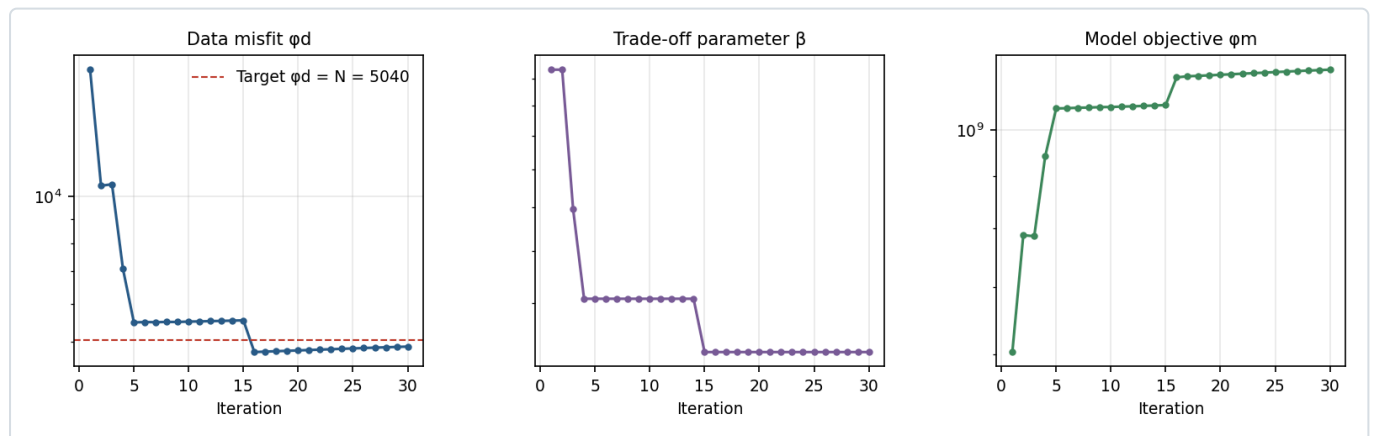


Figure 2 Convergence: data misfit φ_d (red dashed line: target N), trade-off parameter β and model objective φ_m .

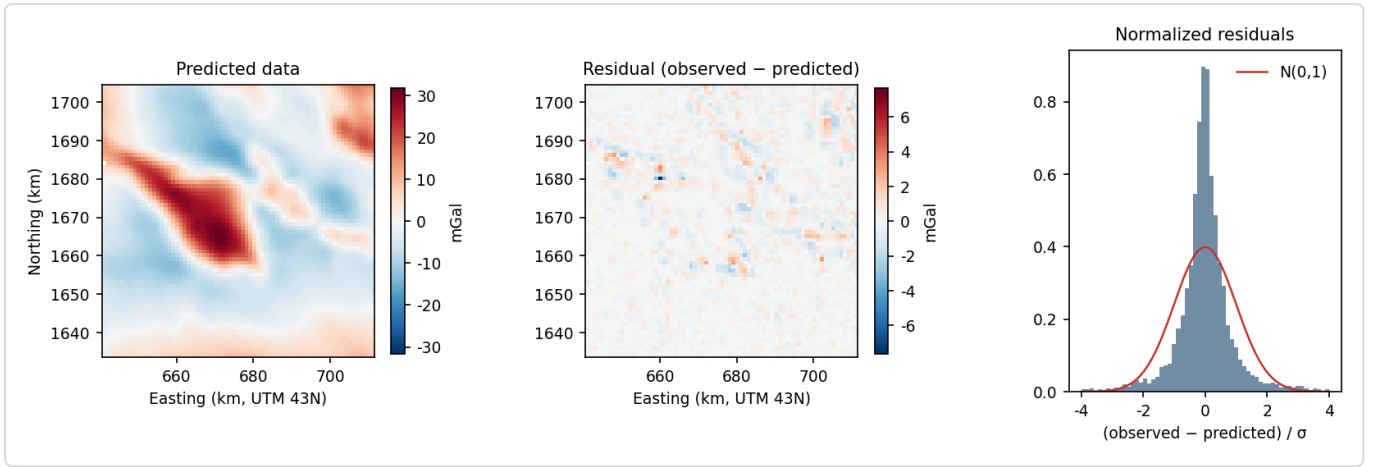


Figure 3 Left: predicted data. Centre: residual (observed – predicted); note the isolated spikes. Right: distribution of normalized residuals against the standard normal $N(0,1)$.

4 The recovered density model

The model reproduces the main structures in the data: a NW–SE dense belt (up to +0.19 g/cc) and several parallel low-density belts to the east (down to -0.16 g/cc). The dense belt follows the regional structural trend and may correspond to a schist or greenstone belt; this should be checked against the geological map.

The problem is the vertical direction. The depth slices at 0.3, 2.3, 5.3 and 9.8 km are almost identical (Figure 4), and at 14.9 km, in the padding, the pattern is the same, only smoother. In the sections (Figure 5) every anomaly is a column from the surface to the bottom of the mesh. The half-maximum extent of the strongest positive anomaly is 0.25–8.3 km, and that of the strongest negative anomaly 0.25–6.8 km.

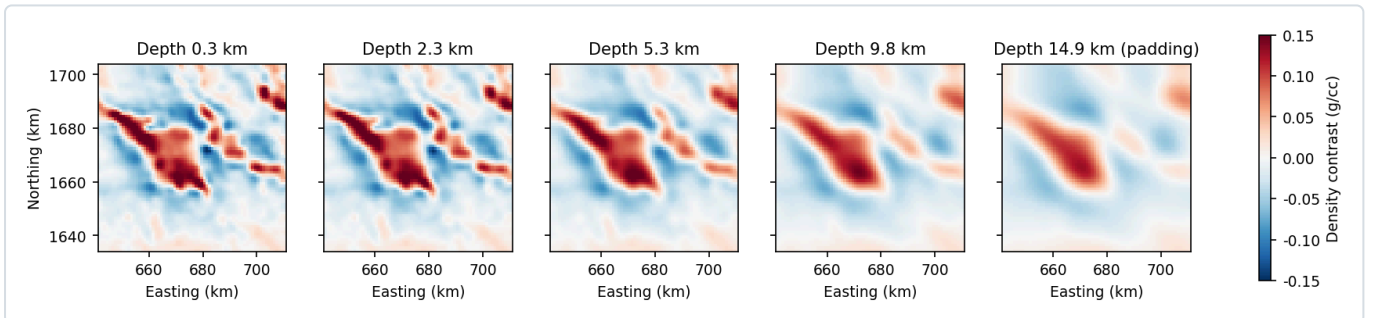


Figure 4 Depth slices of the sparse model (core area only). The first four are nearly identical; the last lies in the padding below 10 km.

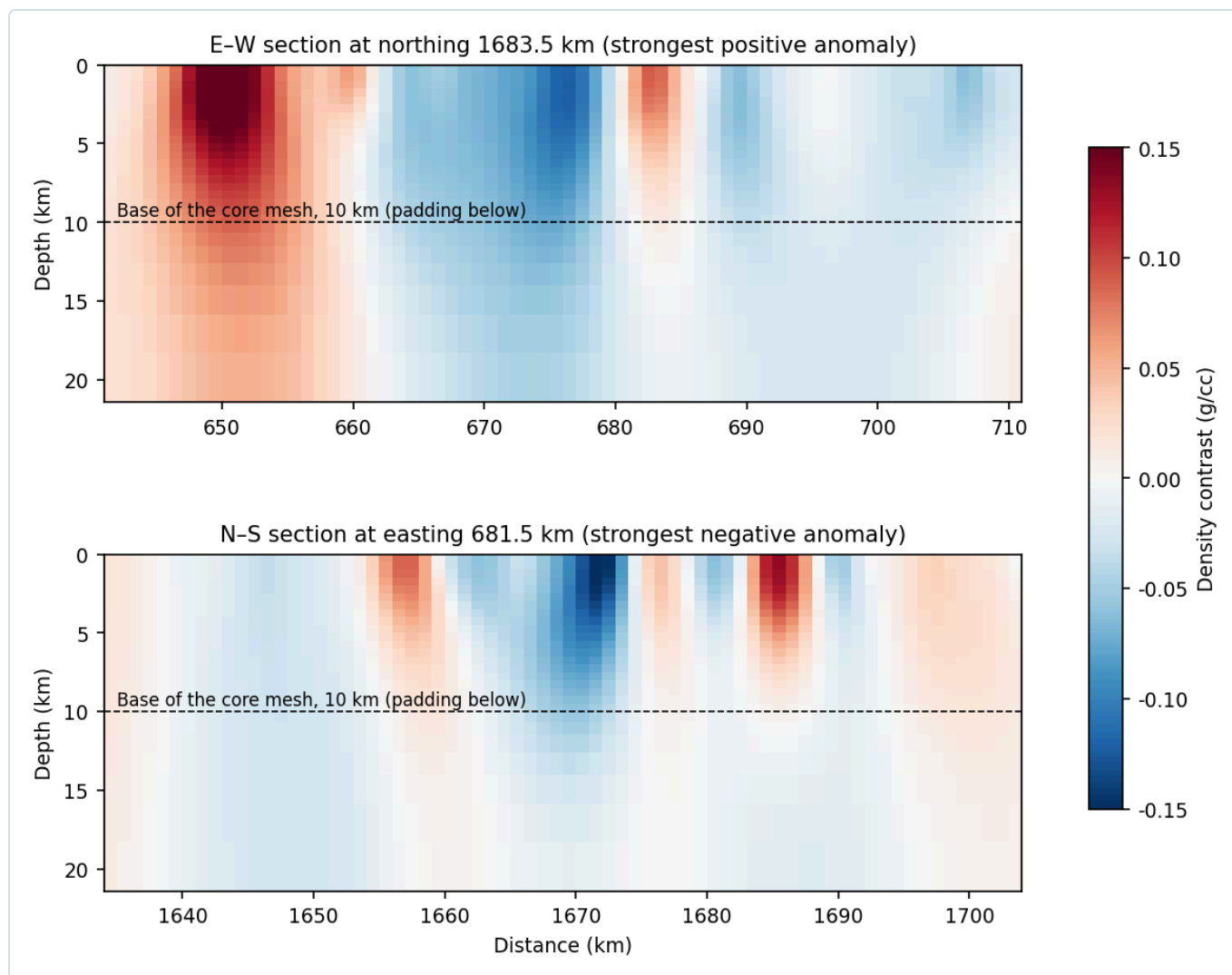


Figure 5 E–W section through the strongest positive anomaly (top) and N–S section through the strongest negative anomaly (bottom). The dashed line is the base of the core mesh (10 km); below it is padding.

5 Why the model extends downward

In numbers

Core / lateral padding / below 10 km	24% / 26% / 50% of density contrast × volume
Depth distribution below the core area	D50 = 9.1 km, D90 = 17.6 km; 72% below 5 km
Spectral depth estimates	deep 6.0 km, intermediate 2.2 km, shallow 1.1 km (Spector & Grant, Figure 6)

Causes

- The compactness term does not act.** The pipeline passes $\alpha_{x,y,z}$ to SimPEG as length scales, so the gradient terms are weighted by $(\text{length scale} \times \text{smallest cell})^2 = (1 \times 500 \text{ m})^2 = 2.5 \times 10^5$. With the $1/h^2$ of the gradient ($h = 1000 \text{ m}$) this is about $0.25 \Delta m^2$ per cell, against $10^{-4} m^2$ for the smallness term, about 2500 times weaker. The sparse (compactness) constraint $p_s = 0$ therefore has almost no effect, and the inversion is in effect driven by the smoothness terms.
- L1 on the vertical gradient.** $p_z = 1$ favours models that are piecewise constant with depth. A column that does not change with depth has zero vertical gradient and costs almost nothing, so it can extend to the bottom of the

mesh.

3. **Sensitivity weighting makes the padding cheap.** The weights follow the data's sensitivity to each cell; padding cells outside the area and at depth have low sensitivity and therefore low weights, so mass placed there costs little. 26% of the mass went into the lateral padding.

Cause 1 dominates: switching only the depth weighting to Li & Oldenburg (`base_dw1`), with α_s still 10^{-4} , gives almost the same result (49% below 10 km), whereas $\alpha_s = 1$ alone brings the share below 10 km down to 22%.

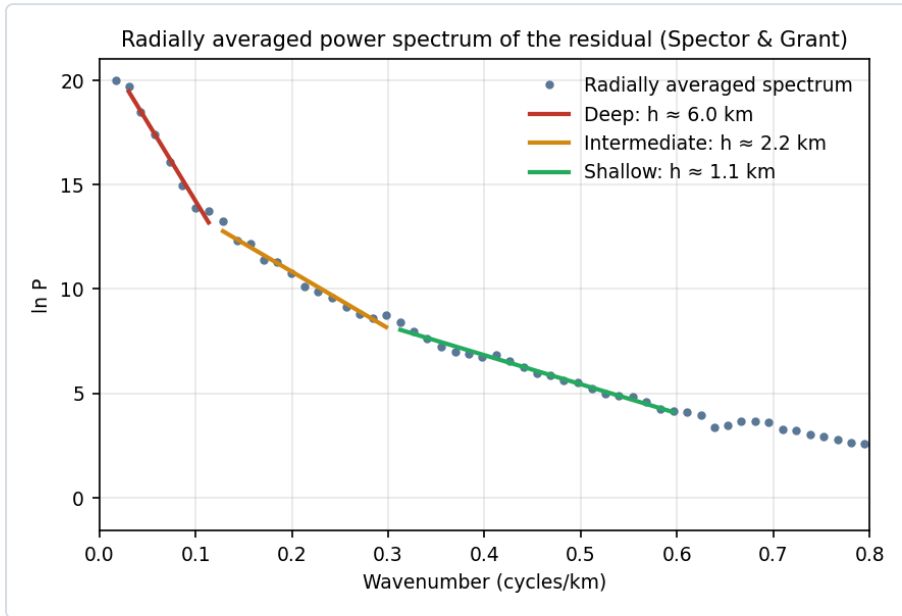


Figure 6 Radially averaged power spectrum of the residual anomaly (500 m grid, second-order trend removed). The slopes of the three line segments give the mean depth to the top of the sources, $h = -\text{slope} / 4\pi$.

6 Parameter study

For a quick comparison, 15 sparse inversions were run locally on a $2 \text{ km} \times 2 \text{ km} \times 1 \text{ km}$ mesh (every fourth point, 1295 data), about 25 s each. The baseline (original settings) reproduces the full-resolution result (D50 8.7 km versus 9.1 km; 50% below 10 km in both), so the coarse runs show the right trends. Depth weighting is the new option `depth_weighting: "depth"`: the model-norm weight is $(z + z_0)^{-\beta/2}$, where z is the depth below the stations and z_0 half the smallest cell.

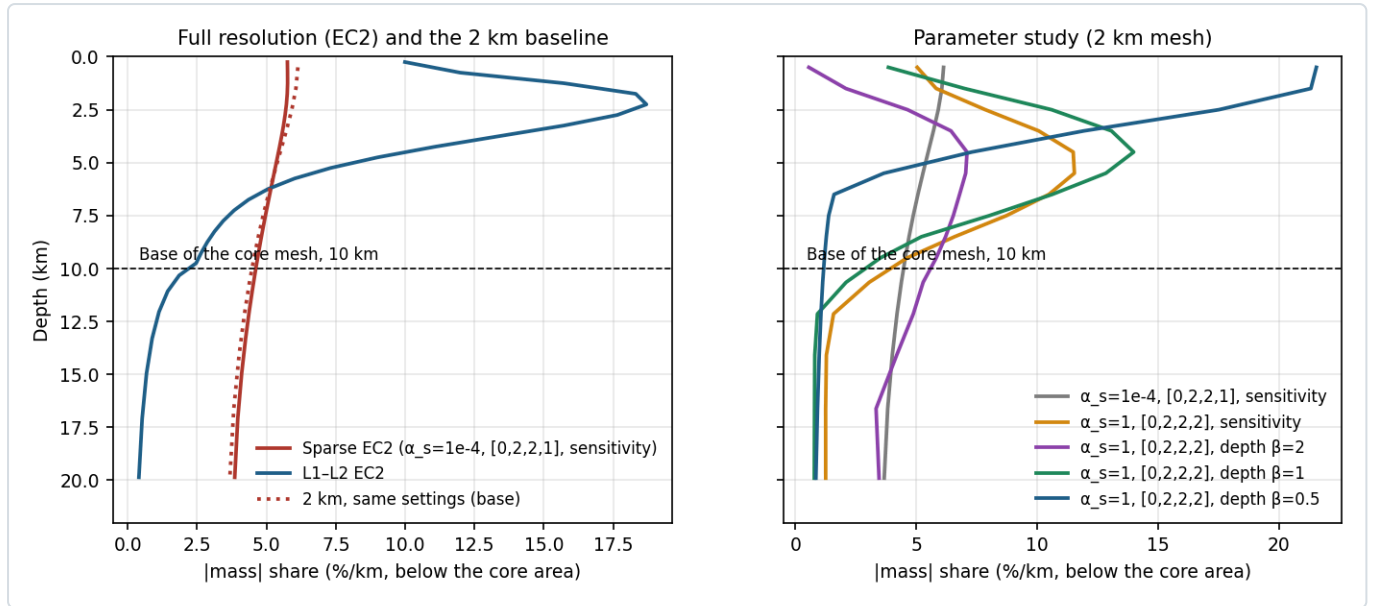


Figure 7 Distribution of |mass| with depth below the core area (share per km). Left: the full-resolution sparse and L1–L2 runs on EC2, with the 2 km baseline. Right: α_s and depth weighting compared on the 2 km mesh.

Run	Mesh	Reg.	α_s	Norms p	Weighting	χ^2/N	Core	Lateral pad.	Below 10 km	D50 km	D90 km	Peak g/cc	Half-max depth km
EC2 sparse (original)	1 km	sparse	0.0001	[0,2,2,1]	sensitivity	0.97	24%	26%	50%	9.1	17.6	+0.19	0.3–8.3
EC2 L1-L2 (IRLS) recommended	1 km	L1–L2	–	–	S1 sensitivity	1.08	80%	9%	11%	3.0	9.2	+0.50	0.8–2.3
base (original)	2 km	sparse	0.0001	[0,2,2,1]	sensitivity	1.00	25%	25%	50%	8.7	17.5	+0.22	0.5–7.5
as1e-2	2 km	sparse	0.01	[0,2,2,1]	sensitivity	1.00	33%	21%	45%	8.2	17.3	+0.23	0.5–6.5
as1	2 km	sparse	1	[0,2,2,1]	sensitivity	1.04	44%	33%	22%	5.9	12.7	+0.50	2.5–7.5
as1_p0111	2 km	sparse	1	[0,1,1,1]	sensitivity	1.09	42%	35%	22%	6.0	13.0	+0.50	3.5–8.5
as1_p0222	2 km	sparse	1	[0,2,2,2]	sensitivity	0.98	47%	33%	20%	5.3	12.9	+0.50	2.5–6.5
as1_p0220	2 km	sparse	1	[0,2,2,0]	sensitivity	0.96	44%	32%	24%	5.8	13.4	+0.50	2.5–4.5
as1e-4_p0222	2 km	sparse	0.0001	[0,2,2,2]	sensitivity	1.08	25%	26%	49%	8.3	17.3	+0.22	0.5–5.5
as1_p0222_dw0.5	2 km	sparse	1	[0,2,2,2]	depth $\beta=0.5$	0.97	80%	5%	15%	1.9	10.5	+0.50	0.5–1.5
as1_p0222_dw1 recommended	2 km	sparse	1	[0,2,2,2]	depth $\beta=1$	1.04	80%	5%	15%	4.6	10.0	+0.50	2.5–6.5
as1_p0222_dw1.5	2 km	sparse	1	[0,2,2,2]	depth $\beta=1.5$	1.00	58%	10%	32%	6.8	14.3	+0.50	3.5–6.5
as1_p0222_dw2	2 km	sparse	1	[0,2,2,2]	depth $\beta=2$	0.94	40%	6%	54%	8.9	17.3	+0.50	2.5–9.5
as1_p0221_dw1	2 km	sparse	1	[0,2,2,1]	depth $\beta=1$	0.94	77%	5%	18%	5.4	10.7	+0.50	2.5–7.5
as1_p0111_dw1	2 km	sparse	1	[0,1,1,1]	depth $\beta=1$	0.99	73%	9%	18%	5.4	10.7	+0.50	3.5–8.5
as0.1_p0222_dw1 recommended	2 km	sparse	0.1	[0,2,2,2]	depth $\beta=1$	1.03	68%	10%	21%	5.3	11.5	+0.29	0.5–4.5
base_dw1	2 km	sparse	0.0001	[0,2,2,1]	depth $\beta=1$	1.03	32%	19%	49%	8.6	17.5	+0.22	0.5–7.5

Core, lateral padding and below 10 km add up to 100%. Red cells: lateral padding > 20%, below 10 km > 30% or D50 > 7 km. The last two columns give the peak of the strongest positive anomaly and its half-maximum depth range; +0.50 means the upper bound was reached.

Observations

- **α_s decides whether the model reaches the bottom.** With $\alpha_s = 10^{-4}$ or 10^{-2} , 45–50% of the mass lies below 10 km; with $\alpha_s = 1$ about 20%.
- **Depth weighting decides the lateral leakage and the mean depth.** With $\alpha_s = 1$ and $p = [0,2,2,2]$, sensitivity weighting leaves 33% in the lateral padding; depth weighting only 5–10%.
- **β sets the mean depth.** $\beta = 0.5 / 1 / 1.5 / 2$ give $D50 = 1.9 / 4.6 / 6.8 / 8.9$ km. The $D50$ for $\beta = 1$ is close to the deep spectral estimate (6.0 km); $\beta = 0.5$ is closer to the intermediate and shallow estimates.
- **The norms matter less.** With $\alpha_s = 1$ and $\beta = 1$, $p = [0,2,2,2]$, $[0,2,2,1]$ and $[0,1,1,1]$ all give $D50$ of 4.6–5.4 km.
- **Bounds.** With $\alpha_s = 1$ the main bodies reach the $+0.5 / -0.2$ g/cc bounds, i.e. blocky and saturated. $\alpha_s = 0.1$ with $\beta = 1$ peaks at $+0.29$ g/cc without saturating, with the anomaly at 0.5–4.5 km, which is a sound compromise.

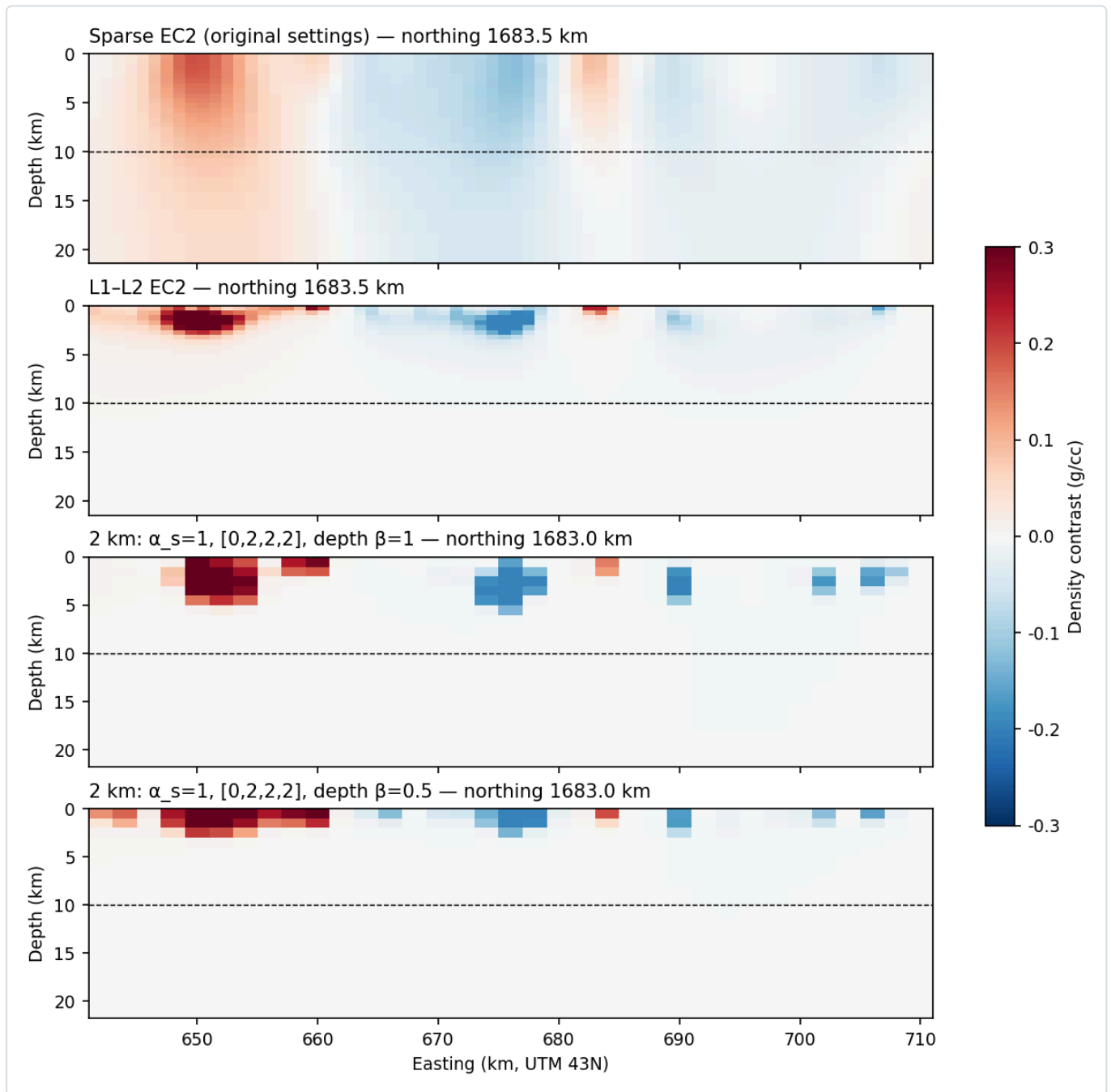


Figure 8 The same E–W section (northing about 1683 km) in four models: the original sparse run, L1–L2, and two depth-weighted settings on the 2 km mesh. Colour scale ± 0.3 g/cc.

7 Full-resolution check

On 28 September four more sparse inversions were run on the same 1 km mesh as the original run (190,512 cells; $p = [0,2,2,2]$, density contrast in $[-0.2, 0.5]$ g/cc, depth weighting `depth`); the $\beta = 0.5$ and 1.5 runs were submitted from the upload page as a parameter sweep. The table also lists the original settings and L1–L2. Depths are read from the density profiles under the main Bouguer high (easting 671.5 km, northing 1664.5 km) and the north-western high (650.5, 1682.5 km).

Run	α_s	Weighting	χ^2/N	Core	Below 10 km	D50 km	Main high: peak depth km	Main high: half-max km	NW high: half-max km
Original sparse ($\alpha_s=1e-4$, sensitivity)	0.0001	sensitivity	0.97	24%	50%	9.1	6.3	0.3–19.9	0.3–10.3
L1–L2 (IRLS)	–	S1	1.08	80%	11%	3.0	3.8	2.3–4.8	1.3–2.8
$\alpha_s=1$, $\beta=0.5$	1	depth $\beta=0.5$	0.90	86%	11%	2.4	3.8	2.3–4.8	0.8–2.8
$\alpha_s=1$, $\beta=1$	1	depth $\beta=1$	0.96	83%	15%	4.6	6.8	3.3–7.8	1.3–4.3
$\alpha_s=1$, $\beta=1.5$	1	depth $\beta=1.5$	0.86	60%	33%	7.1	9.3	4.3–11.1	1.8–5.3
$\alpha_s=0.1$, $\beta=1$	0.1	depth $\beta=1$	0.89	81%	13%	5.4	6.3	3.8–9.3	0.3–4.8

Red cells: below 10 km > 30%, D50 > 7 km, or the main body reaching below 12 km. The peaks reach the +0.5 g/cc upper bound (+0.48 for $\alpha_s = 0.1$).

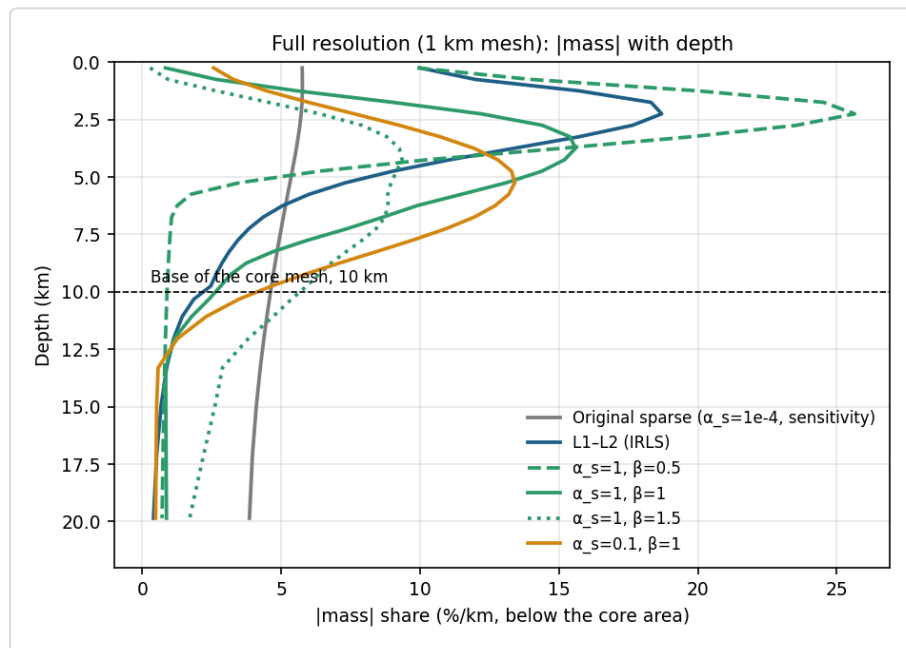


Figure 9 Distribution of $|mass|$ with depth below the core area at full resolution (1 km mesh). Grey: original settings; blue: L1–L2; green: $\alpha_s = 1$ with $\beta = 0.5$ (dashed), 1 (solid) and 1.5 (dotted); orange: $\alpha_s = 0.1$, $\beta = 1$.

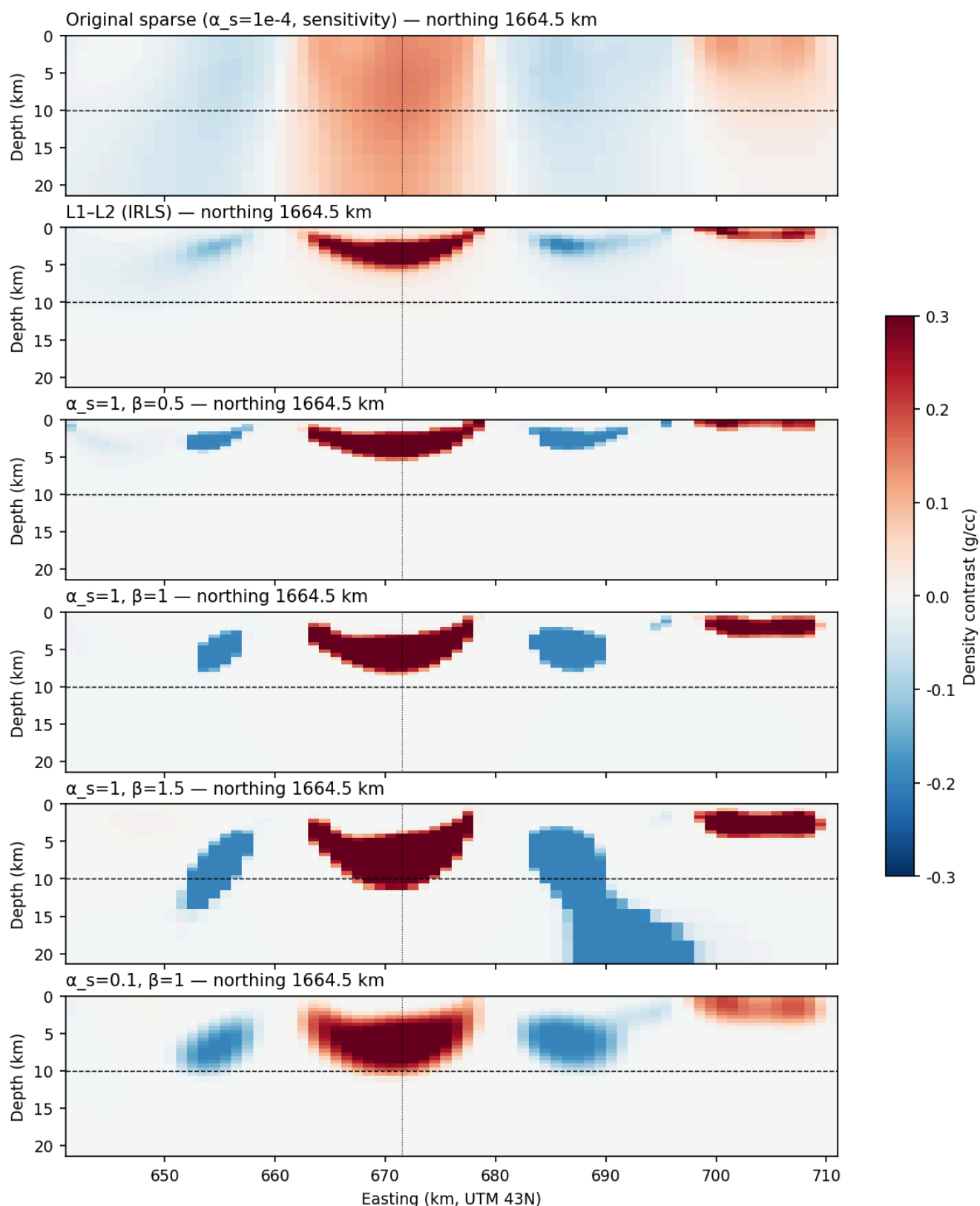


Figure 10 E–W sections through the main Bouguer high (northing 1664.5 km) in the six full-resolution models. The vertical dotted line marks the high, the horizontal dashed line the base of the core mesh. Colour scale ± 0.3 g/cc.

Observations

- **The columns are gone.** With depth weighting the main body ends above 12 km in every run; with the original settings it reached the bottom of the mesh (19.9 km).

- **β sets the depth, and the data cannot tell.** From $\beta = 0.5$ to 1 to 1.5 the peak of the main body moves from 3.8 km to 6.8 km and 9.3 km, while χ^2/N stays between 0.86 and 0.96. This is the non-uniqueness of gravity inversion: choosing β needs independent depth information.
- **$\beta = 0.5$ agrees with L1–L2.** Two quite different regularizations both put the main body at 2.3–4.8 km, near the intermediate spectral estimate (2.2 km); the 6.8 km of $\beta = 1$ is close to the deep estimate (6.0 km).
- **The 2 km study predicted this well.** At the same place the 2 km mesh gave 0.5–4.5 / 3.5–7.5 / 4.5–10.6 km for $\beta = 0.5$ / 1 / 1.5: the base within 0.5 km of the 1 km mesh, the top and the peak 1–3 km shallower.
- **$\beta = 1.5$ starts to trail to depth.** The low-density body in the east reaches 20 km, and 33% of the mass lies below 10 km.

8 Recommendations and next steps

1. **Sparse settings:** $\alpha_s = 1$ (now the pipeline default), $p = [0,2,2,2]$ or $[0,2,2,1]$, depth weighting `depth` with β between 0.5 and 1: $\beta = 0.5$ agrees with L1–L2, $\beta = 1$ with the deep spectral estimate. Report this range of depths rather than a single model.
2. **Fix the depth with independent data:** use rock densities, geological sections, boreholes or seismic data to constrain the top and base of the main dense body, and choose β accordingly. The upload page's parameter sweep submits several β at once and compares them in one workflow.
3. **Handle the spikes in the data:** remove isolated outliers or use a robust misfit, and check whether the 0.5 mGal noise floor is appropriate.
4. **Add prior information:** density bounds by lithology or a reference model; geological sections or seismic data to constrain depth. A joint inversion with the available aeromagnetic TMI data is another option.
5. **Mesh:** the bottom padding cells are thick (up to 3.1 km) and absorb mass. The pipeline's mesh recommendation (2 km cells, 35 km deep core) suits deeper targets; for the upper crust, a 10 km core with depth weighting is enough.
6. **L1–L2 solver:** the coordinate-descent solver of Utsugi (2019) is too slow at 190,000 cells (single-threaded; each λ point took about twice as long as the previous one, 11 of 41 points in 11 minutes). This run used IRLS on the same elastic-net objective and finished in 3.5 minutes.

Files

- All 17 runs in one workflow (DAG viewer):
`examples/output/depth_study/karnataka_depth_study.geoinv3d_viewer.html`
- EC2 results: `~/ .geoinv3d/results/karnataka-gravity-1790537566-sparse` and `...-l112`
- New options: pipeline parameters `depth_weighting` ("sensitivity" or "depth") and `depth_weighting_exponent`; "Depth weighting" on the upload page